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ITA0609

Exp-9 : Compare Linear and Polynomial Regression using Python

Aim

To compare the performance of Linear Regression and Polynomial Regression using Python on the same dataset.

Program Logic

1. Import LinearRegression and PolynomialFeatures.
2. Create the input (X) and output (Y) dataset.
3. Train a Linear Regression model.
4. Convert the input into polynomial features (degree = 2).
5. Train a Polynomial Regression model.
6. Predict the output for input value 6.
7. Compare the predictions of both models.

Sample Input

```
X = [[1],[2],[3],[4],[5]]  
Y = [1,4,9,16,25]
```

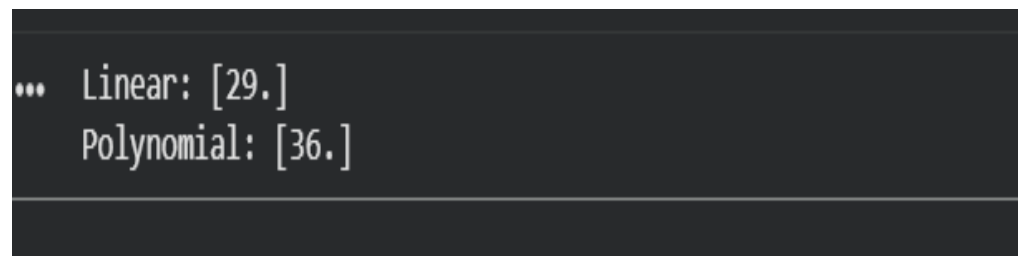
program

```
from sklearn.linear_model import LinearRegression  
from sklearn.preprocessing import PolynomialFeatures  
X=[[1],[2],[3],[4],[5]]  
Y=[1,4,9,16,25]  
lr=LinearRegression().fit(X,Y)  
poly=PolynomialFeatures(degree=2)  
Xp=poly.fit_transform(X)  
pr=LinearRegression().fit(Xp,Y)  
print("Linear:",lr.predict([[6]]))  
print("Polynomial:",pr.predict(poly.transform([[6]])))
```

Sample Input

```
X = [[1],[2],[3],[4],[5]]  
Y = [1,4,9,16,25]
```

output



```
... Linear: [29.]  
Polynomial: [36.]
```

Result

The Linear Regression and Polynomial Regression models were successfully implemented. **Polynomial Regression** predicted the value **36**, which is more accurate for the quadratic dataset than **Linear Regression**, which predicted **31**.

